

## CHATEAU-ARNOUX-ST-AUBAN AP, France

WMO: 075880

Lat: 44.0622N

Lon: 5.9897E

Elev: 1512

StdP: 13.91

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|--------------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                          |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD      |              |
| (a)<br>1      | (b)<br>24.1 | (c)<br>26.6 | (d)<br>8.4                    | (e)<br>8.9 | (f)<br>34.1 | (g)<br>12.1 | (h)<br>10.8 | (i)<br>34.8 | (j)<br>28.9           | (k)<br>44.5 | (l)<br>25.9 | (m)<br>43.0 | (n)<br>6.7               | (o)<br>30 | (p)<br>0.522 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>24.5            | (c)<br>92.2     | (d)<br>66.2 | (e)<br>89.6 | (f)<br>65.7 | (g)<br>87.0 | (h)<br>65.1 | (i)<br>69.0         | (j)<br>85.4 | (k)<br>67.6 | (l)<br>84.1 | (m)<br>66.5 | (n)<br>82.7 | (o)<br>10.8          | (p)<br>210 |

| Dehumidification DP/MCDB and HR |             |             |             |             |             |             |             |             | Enthalpy/MCDB |             |             |             |             |             | Extreme<br>Max WB |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|-------------|-------------|-------------------|
| 0.4%                            |             |             | 1%          |             |             | 2%          |             |             | 0.4%          |             | 1%          |             | 2%          |             |                   |
| DP                              | HR          | MCDB        | DP          | HR          | MCDB        | DP          | HR          | MCDB        | Enth          | MCDB        | Enth        | MCDB        | Enth        | MCDB        |                   |
| (a)<br>63.9                     | (b)<br>94.0 | (c)<br>73.5 | (d)<br>62.3 | (e)<br>89.0 | (f)<br>72.2 | (g)<br>61.0 | (h)<br>84.6 | (i)<br>71.9 | (j)<br>34.0   | (k)<br>85.0 | (l)<br>32.8 | (m)<br>84.2 | (n)<br>31.9 | (o)<br>82.8 | (p)<br>75.6       |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 24.2          | (b) 20.6 | (c) 17.8 | (e) 20.0                   | (f) 96.3 | (g) 3.4            | (h) 3.5 | (i) 17.5                                           | (j) 98.8 | (k) 15.5   | (l) 100.8 | (m) 13.6   | (n) 102.8 | (o) 11.2   | (p) 105.3 |    |    |
| (a) 18.3          | (b) 71.7 | (c) 3.3  | (e) 18.3                   | (f) 71.7 | (g) 3.3            | (h) 1.5 | (i) 15.9                                           | (j) 72.8 | (k) 14.0   | (l) 73.7  | (m) 12.2   | (n) 74.5  | (o) 9.8    | (p) 75.7  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 56.0   | 40.4  | 42.2  | 48.3  | 53.5  | 60.6  | 68.6  | 73.5  | 72.7  | 64.7  | 57.4  | 47.4  | 41.2  |
|                                                                        | DBStd       | 12.84  | 5.47  | 5.64  | 5.61  | 5.35  | 5.35  | 5.72  | 4.48  | 4.49  | 5.04  | 5.35  | 5.94  | 5.46  |
|                                                                        | HDD50       | 1047   | 301   | 222   | 98    | 24    | 2     | 0     | 0     | 0     | 0     | 9     | 116   | 275   |
|                                                                        | HDD65       | 4020   | 763   | 638   | 517   | 344   | 154   | 28    | 4     | 3     | 66    | 238   | 527   | 738   |
|                                                                        | CDD50       | 3226   | 4     | 4     | 46    | 130   | 332   | 559   | 729   | 703   | 440   | 239   | 38    | 2     |
|                                                                        | CDD65       | 724    | 0     | 0     | 0     | 1     | 18    | 136   | 268   | 241   | 56    | 4     | 0     | 0     |
|                                                                        | CDH74       | 7692   | 0     | 0     | 1     | 29    | 266   | 1535  | 2834  | 2417  | 563   | 47    | 0     | 0     |
|                                                                        | CDH80       | 2986   | 0     | 0     | 0     | 1     | 43    | 559   | 1238  | 1017  | 125   | 3     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 7.6    | 7.7   | 8.0   | 8.2   | 7.8   | 7.8   | 7.9   | 8.3   | 7.8   | 7.2   | 6.3   | 7.1   | 7.1   |
| (15) Precipitation                                                     | PrecAvg     | 29.4   | 2.1   | 2.4   | 2.5   | 2.4   | 3.0   | 2.4   | 1.5   | 2.5   | 2.4   | 3.0   | 2.7   | 2.4   |
|                                                                        | PrecMax     | 41.5   | 5.1   | 5.5   | 5.3   | 5.1   | 6.2   | 4.6   | 4.6   | 4.8   | 6.4   | 11.7  | 5.1   | 9.5   |
|                                                                        | PrecMin     | 15.4   | 0.0   | 0.1   | 0.0   | 0.3   | 0.4   | 1.1   | 0.0   | 0.2   | 0.0   | 0.2   | 0.0   | 0.1   |
|                                                                        | PrecStd     | 6.3    | 1.5   | 1.6   | 1.5   | 1.2   | 1.7   | 1.0   | 1.3   | 1.4   | 1.9   | 2.6   | 1.5   | 1.9   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 60.9  | 64.5  | 72.2  | 77.7  | 84.8  | 93.4  | 94.9  | 96.6  | 87.0  | 78.6  | 66.6  | 59.0  |
|                                                                        |             | MCWB   | 49.0  | 48.8  | 52.3  | 58.3  | 63.3  | 67.7  | 66.4  | 66.6  | 64.9  | 60.6  | 54.7  | 50.0  |
|                                                                        | 2%          | DB     | 56.4  | 59.7  | 67.7  | 73.1  | 80.1  | 89.6  | 92.1  | 92.2  | 83.6  | 74.5  | 63.0  | 55.3  |
|                                                                        |             | MCWB   | 46.6  | 46.9  | 50.7  | 56.1  | 60.7  | 65.9  | 66.1  | 65.9  | 63.3  | 59.9  | 52.7  | 46.6  |
|                                                                        | 5%          | DB     | 52.8  | 56.1  | 64.1  | 69.6  | 76.8  | 86.3  | 89.8  | 89.1  | 80.5  | 71.3  | 60.0  | 52.7  |
|                                                                        |             | MCWB   | 44.4  | 45.1  | 49.8  | 54.0  | 59.7  | 65.0  | 65.7  | 65.6  | 62.1  | 58.7  | 51.3  | 45.5  |
|                                                                        | 10%         | DB     | 49.6  | 52.7  | 60.7  | 66.2  | 73.5  | 82.8  | 87.3  | 85.9  | 77.1  | 68.1  | 56.9  | 50.1  |
|                                                                        |             | MCWB   | 42.5  | 43.2  | 47.9  | 52.1  | 58.1  | 63.9  | 65.2  | 64.9  | 60.9  | 57.1  | 50.1  | 44.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 50.8  | 50.5  | 54.7  | 59.4  | 65.3  | 70.8  | 71.2  | 70.7  | 67.4  | 63.7  | 57.1  | 51.7  |
|                                                                        |             | MCDB   | 57.5  | 61.3  | 66.5  | 73.9  | 79.2  | 88.5  | 87.3  | 86.1  | 80.3  | 73.1  | 61.2  | 55.9  |
|                                                                        | 2%          | WB     | 48.0  | 48.1  | 52.6  | 57.3  | 62.9  | 67.9  | 69.1  | 68.5  | 65.6  | 61.6  | 54.8  | 49.2  |
|                                                                        |             | MCDB   | 54.4  | 56.7  | 63.9  | 70.6  | 76.9  | 85.0  | 86.1  | 84.5  | 78.8  | 70.9  | 60.1  | 53.0  |
|                                                                        | 5%          | WB     | 45.7  | 46.2  | 50.8  | 55.3  | 61.0  | 66.3  | 67.7  | 67.3  | 64.1  | 60.0  | 52.8  | 46.9  |
|                                                                        |             | MCDB   | 51.4  | 54.0  | 61.6  | 67.1  | 73.5  | 82.5  | 84.8  | 83.6  | 76.6  | 68.6  | 57.9  | 51.0  |
|                                                                        | 10%         | WB     | 43.4  | 44.3  | 49.1  | 53.4  | 59.3  | 64.8  | 66.5  | 66.1  | 62.4  | 58.3  | 51.1  | 44.8  |
|                                                                        |             | MCDB   | 48.3  | 51.3  | 58.9  | 63.7  | 70.9  | 80.0  | 83.1  | 82.3  | 73.3  | 65.7  | 56.0  | 48.7  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 15.6  | 18.6  | 21.1  | 21.0  | 21.4  | 23.4  | 24.5  | 24.1  | 21.6  | 19.2  | 15.9  | 14.8  |
|                                                                        |             | MCDBR  | 19.8  | 24.6  | 27.0  | 26.6  | 26.8  | 28.7  | 29.1  | 29.5  | 26.8  | 23.6  | 19.9  | 18.1  |
|                                                                        |             | MCWBR  | 12.4  | 14.3  | 14.0  | 12.8  | 11.6  | 10.5  | 9.9   | 9.9   | 10.4  | 11.6  | 11.6  | 11.3  |
|                                                                        | 5% WB       | MCDBR  | 17.4  | 22.1  | 24.2  | 24.8  | 24.3  | 26.5  | 26.5  | 26.2  | 23.7  | 20.7  | 16.6  | 14.9  |
|                                                                        |             | MCWBR  | 11.8  | 13.8  | 13.3  | 12.6  | 11.3  | 10.6  | 9.9   | 9.9   | 10.1  | 10.9  | 10.6  | 10.4  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.283 | 0.303 | 0.338 | 0.378 | 0.390 | 0.392 | 0.382 | 0.375 | 0.365 | 0.345 | 0.311 | 0.282 |
|                                                                        | taud        |        | 2.513 | 2.472 | 2.418 | 2.319 | 2.331 | 2.354 | 2.380 | 2.400 | 2.418 | 2.482 | 2.519 | 2.530 |
|                                                                        | Ebn at Noon |        | 272   | 284   | 285   | 281   | 280   | 279   | 280   | 279   | 272   | 264   | 258   | 261   |
|                                                                        | Edh at Noon |        | 22    | 27    | 32    | 38    | 39    | 38    | 37    | 35    | 32    | 26    | 22    | 19    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 573   | 866   | 1259  | 1594  | 1888  | 2202  | 2241  | 1942  | 1475  | 954   | 599   | 491   |
|                                                                        | RadStd      |        | 63    | 93    | 126   | 134   | 154   | 113   | 122   | 80    | 99    | 91    | 79    | 54    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | +0.65 | N/A     | N/A    | +1.85   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (35 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page